Project report

on

Air Quality Index (AQI) Prediction

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This project has been a valuable learning experience, enhancing my knowledge of data processing, machine learning, and real-world problem-solving.

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Declaration

I hereby declare that the project titled “Air Quality Index (AQI) Prediction using Machine Learning in PySpark” is an original work carried out by me. This project has been completed under the guidance of my faculty and has not been submitted previously, either in part or in full, to any other institution for the award of any degree or certification.

All sources of information used in this project have been properly acknowledged.

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Introduction

Air pollution has become a major global concern, significantly affecting human health, environmental sustainability, and quality of life. Rapid urbanization, industrialization, and increased vehicular emissions have led to a rise in harmful pollutants such as particulate matter (PM2.5 and PM10), nitrogen dioxide (NO_2), sulfur dioxide (SO_2), carbon monoxide (CO), and ozone (O_3). These pollutants directly influence the Air Quality Index (AQI), which is widely used as an indicator to assess and communicate air pollution levels.

This project focuses on predicting the Air Quality Index (AQI) using environmental and atmospheric data collected from multiple cities across the world. The dataset includes key pollutant concentrations along with meteorological factors such as temperature and humidity. By leveraging big data processing capabilities of PySpark and applying machine learning techniques, the project aims to build a predictive model that can estimate AQI levels based on these features.

A Linear Regression model is implemented to analyze the relationship between pollutant concentrations and AQI values. Data preprocessing steps such as handling missing values, feature transformation, and normalization are performed to ensure model accuracy and efficiency. The performance of the model is evaluated using metrics such as Root Mean Squared Error (RMSE) and R^2 score.

The outcome of this project provides insights into how different environmental factors contribute to air pollution and demonstrates the effectiveness of machine learning in predicting air quality. Such predictive models can be useful for environmental monitoring agencies and policymakers to take proactive measures in controlling pollution and improving public health.

Objectives

The primary objective of this project is to develop a machine learning model to predict the Air Quality Index (AQI) based on environmental and pollutant data. The project aims to analyse how different atmospheric factors influence air quality and to provide accurate predictions using data-driven techniques.

The specific objectives of the project are as follows:

* To load and process air quality data using PySpark for efficient handling of large datasets.
* To clean the dataset by handling missing values and ensuring data consistency.
* To perform feature engineering by selecting relevant pollutant and environmental variables such as PM2.5, PM10, NO_2, CO, O_3, temperature, and humidity.
* To normalise numerical features to improve model performance and stability.
* To implement a Linear Regression model for predicting AQI values.
* To evaluate the performance of the model using metrics such as Root Mean Squared Error (RMSE) and R^2 score.
* To visualise and compare actual and predicted AQI values for better understanding of model accuracy.
* To analyse the relationship between different pollutants and their impact on air quality.

The overall goal is to demonstrate the application of big data analytics and machine learning techniques in solving real-world environmental problems.

Dataset Description

Source of Data: **Air Quality Dataset**

The dataset used in this project is a global air quality dataset that contains environmental and pollutant measurements collected from multiple cities across different countries. The dataset is designed to capture variations in air pollution levels over time and across regions, making it suitable for predictive analysis of the Air Quality Index (AQI).

The dataset consists of 1000 records, with each record representing air quality measurements for a specific city on a particular date. It includes a combination of pollutant concentration values and meteorological parameters that influence air quality.

Attributes of the Dataset

Date: Represents the date on which the air quality measurements were recorded. This helps in tracking temporal variations in pollution levels.
City (Region): Indicates the geographical location where the data was collected. It includes major cities from different parts of the world such as Delhi, New York, London, Beijing, Tokyo, and others.
PM2.5 (Particulate Matter ≤ 2.5 µm): Fine particulate matter that can penetrate deep into the lungs and is a major contributor to air pollution.
PM10 (Particulate Matter ≤ 10 µm): Coarser particles that affect respiratory health and visibility.
NO_2 (Nitrogen Dioxide): A harmful gas primarily emitted from vehicles and industrial activities.
SO_2 (Sulfur Dioxide): Emitted from fossil fuel combustion and industrial processes.
CO (Carbon Monoxide): A toxic gas produced by incomplete combustion of fuels.
O_3 (Ozone): A secondary pollutant formed by chemical reactions in the atmosphere under sunlight.
Temperature: Represents atmospheric temperature, which affects chemical reactions and pollutant dispersion.
Humidity: Indicates moisture content in the air, influencing pollutant concentration and air quality.
Wind Speed: A meteorological factor that affects the dispersion of pollutants in the atmosphere.

Derived Attribute

AQI (Air Quality Index): Since the dataset does not originally contain AQI values, it is computed using the average of key pollutant concentrations such as PM2.5, PM10, NO_2, SO_2, CO, and O_3. This derived attribute serves as the target variable for the prediction model.

Characteristics of the Dataset

* The dataset contains both numerical and categorical variables, requiring preprocessing techniques such as encoding and normalisation.
* It includes noisy and potentially missing values, which are handled during the data cleaning phase.
* The dataset represents real-world environmental conditions, making it suitable for machine learning applications.
* It captures both spatial (city-wise) and temporal (date-wise) variations in air quality.

This dataset provides a comprehensive view of environmental pollution factors and serves as a strong foundation for building predictive models to estimate AQI levels using machine learning techniques.

Implementation Using PySpark

Step 1: Importing Required Libraries

- The statement 'from pyspark.sql import SparkSession'(syntax) is used to import the SparkSession class from the PySpark SQL module.
- SparkSession is the primary entry point for working with structured data in PySpark.*
- It allows users to create DataFrames, execute SQL queries, and manage the overall Spark application.
- This import is essential as all subsequent data processing and machine learning operations depend on the Spark session.

Step 2: Loading the Dataset

- The application is named "AQI_Prediction" to identify the workflow during execution.
- The air quality dataset is loaded from a CSV file into a Spark DataFrame using the `spark.read.csv()` function.
- The parameters `header=True` and `inferSchema=True` are used to ensure that the first row is treated as column names and that data types are automatically detected by Spark.
- The `show()` function is used to display a sample of the dataset, allowing quick verification of data loading.
- The `printSchema()` function is used to examine the structure of the dataset, including column names and their respective data types.
- This step ensures that the dataset is correctly loaded and understood before proceeding to data cleaning and preprocessing.

Step 3: Data Cleaning and Pre-Processing

- The dataset initially contains column names with special characters, such as "PM2.5", which can cause issues during processing in PySpark.
- To ensure compatibility, the column is renamed to "PM2_5" using the 'withColumnRenamed()' function. This makes the column name easier to reference in subsequent operations.
- Missing or null values in the dataset are handled using the 'dropna()' function.
- Instead of removing all rows with any null value, the operation is specifically applied to important pollutant columns such as PM2_5, PM10, NO2, CO, and O3.
- This ensures that only rows with missing critical environmental data are removed, while preserving as much useful data as possible.

- Handling missing values is essential to improve the accuracy and reliability of the machine learning model.
- This step prepares the dataset for further preprocessing and feature engineering.

Step 4: Converting Categorical Variables

- The dataset contains categorical data in the form of the city (or region) column, which represents different geographical locations.
- Machine learning algorithms in PySpark require numerical input, so categorical variables must be converted into numeric form.
- The 'StringIndexer' class from 'pyspark.ml.feature' is used to perform this conversion.
- The 'inputCol' parameter specifies the original categorical column (city), and the 'outputCol' parameter defines the new numerical column (city_index).
- The 'fit()' method analyses the unique categories in the dataset and assigns a numerical index to each city.
- The 'transform()' method applies this mapping to the dataset, creating a new column with numerical representations of the categorical values.
- This step enables the inclusion of geographical information in the machine learning model while maintaining compatibility with numerical processing requirements.

Step 5: Feature Selection

- In this step, relevant input variables (features) are selected for building the machine learning model.
- The selected features include key pollutant concentrations such as CO, PM2.5 (PM2_5), PM10, NO2, and O3, which directly influence air quality.
- Additionally, the encoded categorical variable 'city_index' is included to incorporate geographical variation in pollution levels.
- These features are stored in a list named 'feature_columns', which will be used in subsequent steps for feature transformation.
- Selecting appropriate features is crucial, as it directly impacts the accuracy and performance of the predictive model.
- This step ensures that only meaningful and relevant variables are used for AQI prediction, avoiding unnecessary complexity.

Step 6: Feature Vectorisation and Normalisation

- In this step, the selected features are combined into a single vector to prepare the data for machine learning algorithms.

- The ‘VectorAssembler’ class is used to merge multiple feature columns into one column named features_unscaled.
- This transformation is necessary because PySpark ML models require input data in vector format rather than separate columns.
- The ‘transform()’ function is applied to convert the dataset into this structured format.
- After assembling the features, normalisation is performed using the ‘StandardScaler’ class.
- Normalisation scales the feature values so that they have a consistent range, reducing the impact of large variations between different variables.
- The scaler takes features_unscaled as input and produces a normalised output column named features.
- This step improves model performance and ensures that all features contribute equally during training.
- The transformed dataset is stored in the variable data, which is used for further modeling steps.

Step 7: Creation of Target Variable (AQI)

- The original dataset does not contain a predefined Air Quality Index (AQI) column, which is required as the target variable for prediction.
- To address this, a new column named AQI is created using the ‘withColumn()' function.
- The AQI value is calculated as the average of major pollutant concentrations, including PM2.5, PM10, NO2, SO2, CO, and O3.
- The ‘col()’ function is used to reference individual columns and perform arithmetic operations on them.
- This derived AQI serves as an approximate indicator of overall air quality for each record.
- Creating the AQI column enables the application of supervised machine learning techniques, where AQI acts as the dependent (target) variable.
- This step is crucial for transforming raw environmental data into a format suitable for predictive modeling.

Step 8: Feature Transformation and Final Dataset Preparation

- In this step, the selected feature columns are transformed into a format suitable for machine learning algorithms.
- The ‘VectorAssembler’ class is used to combine multiple input features into a single vector column named features_unscaled.
- This transformation is necessary because PySpark ML models require input data to be in vector form rather than separate columns.
- The ‘transform()’ method is applied to generate this new feature vector from the dataset.
- After assembling the features, normalisation is performed using the `StandardScaler` class.
- The scaler standardises the values of the feature vector, ensuring that all variables

contribute equally and reducing the impact of differences in scale.

- The normalised feature vector is stored in a new column named features.
- Finally, a new DataFrame named final_data is created by selecting only the required columns: a. features (independent variables), b.AQI (target variable)
- This final dataset serves as the input for training and testing the machine learning model.

Step 9: Splitting the Dataset into Training and Testing Sets

- In this step, the prepared dataset is divided into two subsets: training data and testing data.
- The 'randomSplit()' function is used to split the dataset in an 80:20 ratio.
- Approximately 80% of the data is used as the training set (train_data) to train the machine learning model.
- The remaining 20% of the data is used as the testing set (test_data) to evaluate the model's performance on unseen data.
- This random splitting ensures that the data is distributed without bias, improving the reliability of the model evaluation.
- Separating the dataset into training and testing sets helps prevent overfitting and ensures that the model generalises well to new data.

Step 10: Training the Linear Regression Model

- In this step, a Linear Regression algorithm is used to build a predictive model for estimating AQI values.
- The `LinearRegression` class from 'pyspark.ml.regression' is imported to implement the model.
- The model is configured by specifying:
- a. featuresCol = "features"** → input variables (independent features)
- labelCol = "AQI"** → target variable (dependent variable)
- The `fit()` method is applied to the training dataset (**train_data**) to train the model.
- During training, the model learns the relationship between pollutant concentrations and AQI values.
- The trained model is stored in the variable **model**, which will be used for making predictions.
- This step is crucial as it enables the system to estimate AQI based on environmental factors using learned patterns from the data.

Step 11: Model Prediction

- In this step, the trained Linear Regression model is used to make predictions on the testing dataset.
- The 'transform()' method is applied to the test_data, generating predicted AQI values based on the learned model.
- The output is stored in a new DataFrame called predictions, which includes both actual and predicted values.
- A new column named prediction is automatically created by the model, representing the estimated AQI values.
- The 'select()' function is used to display only the relevant columns, namely AQI (actual values) and prediction (predicted values).
- The 'show()' function is used to visualize a sample of the results for quick comparison.
- This step helps in understanding how well the model performs by comparing predicted AQI values with the actual values.

Step 12: Model Evaluation

- In this step, the performance of the trained Linear Regression model is evaluated using standard regression metrics.
- The 'RegressionEvaluator' class from 'pyspark.ml.evaluation' is used to measure how accurately the model predicts AQI values.
- The Root Mean Squared Error (RMSE) is calculated to measure the average difference between actual and predicted AQI values. A lower RMSE indicates better model performance.
- The R^2 (R-squared) score is calculated to determine how well the model explains the variance in the data. A value closer to 1 indicates a better fit.
- The 'evaluate()' function is used to compute these metrics based on the predictions DataFrame.
- The results are printed using the 'print()' function for easy interpretation.
- This step is essential for assessing the accuracy and reliability of the model before drawing conclusions.

Step 13: Visualisation of Results

- In this step, the model's performance is visualised by comparing actual and predicted AQI values.
- The Spark DataFrame is converted into a Pandas DataFrame using the 'toPandas()' function to enable plotting with visualisation libraries.
- The data is sorted based on the AQI values to create a smoother and more interpretable graphical representation.

- The ‘matplotlib’ library is used to generate a line plot for visualisation.
- Two lines are plotted: Actual AQI values V/S Predicted AQI values
- Labels for the x-axis, y-axis, and title are added to make the graph informative and easy to understand.
- A legend is included to distinguish between actual and predicted values.
- The ‘show()’ function is used to display the final visualisation.
- This step helps in visually analysing how closely the predicted values match the actual AQI values, providing insight into the model’s accuracy.

Visual representation

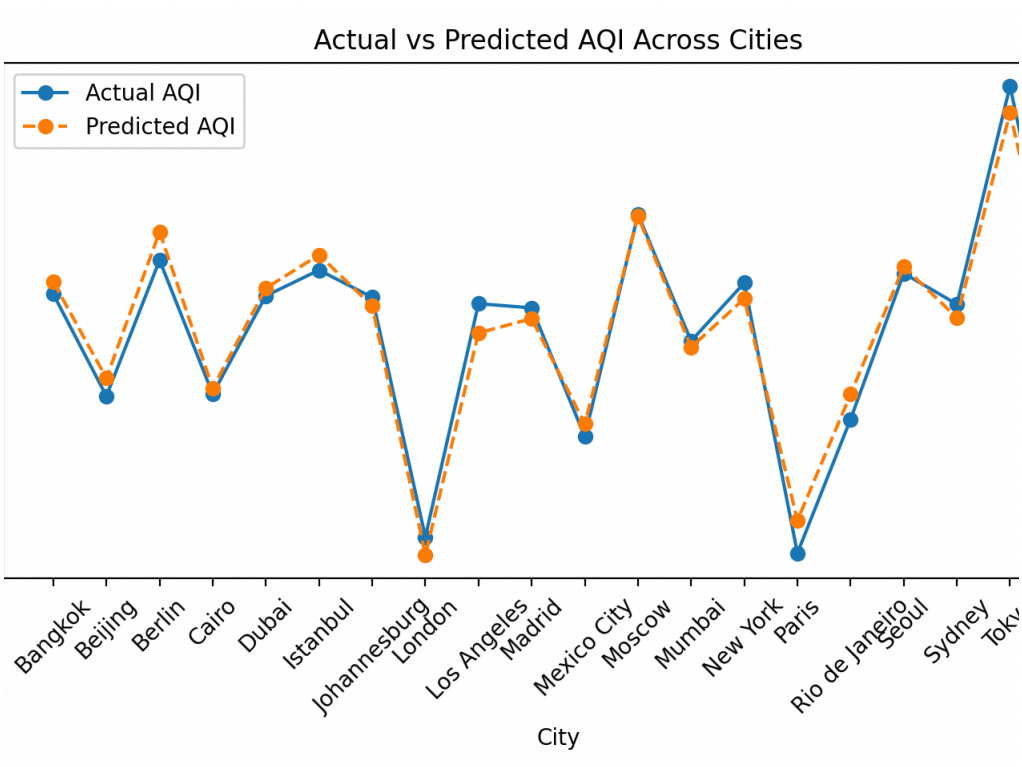

Figure: Actual AQI v/s Predicted AQI, City Wise

Results and Analysis

The Linear Regression model was successfully trained to predict Air Quality Index (AQI) values using environmental and pollutant features. The model was evaluated using standard regression metrics, and the following results were obtained:

Root Mean Squared Error (RMSE): 2.38
R^2 Score:0.978

A comparison between actual and predicted AQI values shows that the predicted values are very close to the actual values. For example:

* Actual AQI: 63.36 → Predicted AQI: 63.48
* Actual AQI: 82.88 → Predicted AQI: 83.21
* Actual AQI: 97.27 → Predicted AQI: 96.01

The visualisation of actual vs predicted AQI values also shows that both curves almost overlap, indicating high prediction accuracy.

The results indicate that the Linear Regression model performs exceptionally well in predicting AQI values:

* The low RMSE (2.38) suggests that the average prediction error is minimal, meaning the model's predictions are very close to the actual AQI values.
* The high R^2 value (0.978) indicates that approximately 97.8% of the variance in AQI is explained by the model, demonstrating a strong relationship between the selected features and AQI.
* The close alignment between actual and predicted values confirms that the model has effectively learned the patterns in the dataset.
* The overlapping curves in the visualisation further validate the model's accuracy and consistency.

However, minor differences between actual and predicted values can still be observed in some cases. These variations may be due to:

* Simplified AQI calculation (derived as an average of pollutants)
* Possible non-linear relationships between variables
* Real-world variability in environmental condition.

Conclusion

This project successfully demonstrates the use of machine learning techniques for predicting Air Quality Index (AQI) using environmental and pollutant data. The use of PySpark enabled efficient processing and analysis of the dataset.

A Linear Regression model was implemented to analyze the relationship between pollutant concentrations (PM2.5, PM10, NO_2, SO_2, CO, O_3) and AQI values.

The model was evaluated using performance metrics, where the obtained results were:

* RMSE = 2.38, indicating a very low prediction error
* $R^2 = 0.978$, indicating that approximately 97.8% of the variance in AQI is explained by the model.

The predicted AQI values were very close to the actual values, as observed in the output. For example:

63.36 → 63.48

82.88 → 83.21

97.27 → 96.01

These results demonstrate high prediction accuracy.

The visualization of actual versus predicted AQI showed overlapping curves, further confirming that the model performs accurately and consistently.

However, the AQI values were derived using a simplified averaging method rather than standard AQI calculation formulas, which may limit real-world accuracy. Additionally, Linear Regression assumes a linear relationship, whereas environmental data may exhibit more complex patterns.

GITHUB LINK: https://github.com/Jayk-2004/AQI-Prediction-/tree/main

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